The Effective Shafarevich Conjecture
Thesis summary

Robin Visser

University of Warwick; supervisor Samir Siksek.
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Overview. Let K be a number field, d a positive integer and S a finite set of primes of K. Faltings proved that there are only finitely many abelian varieties A/K of dimension d with good reduction away from S. Whilst this is considered one of the crowning achievements of 20th century arithmetic geometry, his proof is not fully effective. Though many algorithms list elliptic curves with good reduction outside S, in higher dimension there is in general no algorithm that outputs all such abelian varieties. Finding one is the *effective Shafarevich problem*. The thesis develops theoretical and computational tools for this problem, focusing on hyperelliptic curves and on the case $K = \mathbb{Q}$, $d = 2$, $S = \{2\}$, a question posed by Poonen in 1996. Chapter 2 has appeared as a preprint, Chapter 3 is joint work with Siksek published in *Algebra & Number Theory* (2025), and Chapters 4–6 are new unpublished work.

Survey (Chapter 1). The first chapter surveys the known methods, starting with the many algorithms for elliptic curves (from Ogg's list of the 24 elliptic curves over $\mathbb{Q}$ with good reduction outside 2 to modern S-unit and Thue–Mahler approaches), and the far fewer results for higher genus curves and higher-dimensional abelian varieties. It also sets up the tools used later in the thesis: models of hyperelliptic curves, Jacobians and their torsion fields, cluster pictures, invariants of genus 2 curves, L-functions, and modularity results and conjectures.

Potential good reduction of hyperelliptic curves (Chapter 2). Let C/K be a hyperelliptic curve of genus g with all Weierstrass points in K. Using cluster pictures, I show that C has potentially bad reduction at every odd prime of norm less than $2g$, and that only finitely many such curves have potentially good reduction outside $\pi_{K,\mathrm{odd}}(2g) + 1$ primes. Over $\mathbb{Q}$ this extends a result of Box and Le Fourn: no genus 2 curve with rational Weierstrass points has potentially good reduction outside one prime, and no genus 3 curve outside two primes. Writing $c_K(g)$ for the least number of primes such that infinitely many such curves have potentially good reduction outside that many primes, I prove the unconditional upper bound $c_K(g) \leq \left(\frac{2}{\log 2} + o(1)\right) ng \log g$ for K of degree n, together with sharper bounds under the Hardy–Littlewood prime k-tuples conjecture and Schinzel's Hypothesis H. In contrast, the Jacobian behaves differently from the curve: there are infinitely many genus 2 and genus 3 curves over $\mathbb{Q}$ with rational Weierstrass points whose Jacobians have potentially good reduction outside $\{2\}$.

Curves over $\mathbb{Z}_\ell$-cyclotomic extensions (Chapter 3; with Siksek). Conjectures of Mazur and of Parshin–Zarhin, and a theorem of Zarhin, suggest that curves and abelian varieties over the $\mathbb{Z}_\ell$-cyclotomic extension $\mathbb{Q}_{\infty,\ell}$ behave much as over number fields. We show that the Siegel–Mahler and Shafarevich finiteness theorems nevertheless fail there. Using identities between cyclotomic polynomials, we prove constructively that $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ has infinitely many S-integral points over $\mathbb{Q}_{\infty,\ell}$ for $\ell = 2, 3, 5, 7$, where S contains at most the prime above ℓ. From these we build infinitely many elliptic curves over $\mathbb{Q}_{\infty,\ell}$ with full 2-torsion and good reduction away from the primes above 2 and ℓ, falling into infinitely many isogeny classes. This suggests that the Parshin–Zarhin conjecture cannot be proved by imitating Faltings' proof of the Mordell conjecture. Reassuringly, using a theorem of Kato, each isogeny class of elliptic curves without potential CM is finite. Using Kummer's relation between class numbers and cyclotomic units, we also construct infinite families of hyperelliptic curves of every genus $g \geq 2$ (for $\ell = 3, 5, 7, 11, 13$) with good reduction away from 2 and ℓ.

Abelian surfaces with full rational 2-torsion (Chapter 4). A principally polarised abelian surface over $\mathbb{Q}$ is a Jacobian of a genus 2 curve, a product of elliptic curves, or a Weil restriction of an elliptic curve over a quadratic field. The last two cases with good reduction away from 2 reduce to known classifications of elliptic curves, so the problem comes down to Poonen's question: list all genus 2 curves over $\mathbb{Q}$ whose Jacobians have good reduction away from 2. As a first step, I prove that there are exactly three isogeny classes of principally polarised abelian surfaces over $\mathbb{Q}$ with good reduction away from 2 that contain a surface with full rational 2-torsion: those of $E_1 \times E_1$, $E_1 \times E_2$ and $E_2 \times E_2$, where $E_1 : y^2 = x^3 - x$ and $E_2 : y^2 = x^3 - 4x$. The proof avoids explicit models of abelian surfaces. For such a surface, $\mathbb{Q}(A[2])$ is a 2-extension of degree at most 2^6 unramified outside 2. Using the Faltings–Serre method, with deviation groups and the criteria of Livné and Grenié, I show that the isogeny class is determined by the Euler factors at $p = 5, 7, 11, 17, 23, 31$. I then determine all possible Euler factors by enumerating the possible Galois images in $\mathrm{GSp}_4(\mathbb{Z}/64\mathbb{Z})$, which requires an efficient solution of the subgroup conjugacy problem in these groups. I conjecture that the three products are in fact the only such surfaces up to isomorphism.

Computing abelian surfaces with good reduction outside 2 (Chapter 5). Here I develop three complementary approaches to Poonen's question. (i) *L-functions.* Extending work of Farmer, Koutsoliotas and Lemurell, I show, assuming the paramodular conjecture, that there are no abelian surfaces over $\mathbb{Q}$ of conductor 2^n for $n \leq 7$ or $n = 9$, exactly one isogeny class of conductor 2^8, and exactly one of conductor 2^{10} containing a surface with $(\mathbb{Z}/2\mathbb{Z})^3 \subseteq A(\mathbb{Q})$. (ii) *Curves with good reduction outside a set S.* Extending Smart's method, I classify the possible 2-torsion fields (number fields of degree at most 6 unramified outside 2) and write each root difference $\alpha_i - \alpha_j$ in a basis of S-units. Galois symmetries, cluster pictures at odd primes and solutions of S-unit equations give linear constraints on the exponents, and the resulting systems are solved with closest vector and integer linear programming techniques. This allows curves with bad reduction at odd primes, for S containing primes up to 13. (iii) *Gluing.* I construct genus 2 curves by gluing pairs of elliptic curves with good reduction away from 2 along their n-torsion, following Frey and Kani, starting from the 55 isogeny classes of such products. Together these show that there are at least 234 isogeny classes of abelian surfaces over $\mathbb{Q}$ with good reduction away from 2, including 512 genus 2 curves whose Jacobians have good reduction away from 2, in 175 isogeny classes. The list is provably complete within each of eleven explicit families, for example curves with good reduction outside $\{2, 3\}$, $\{2, 5\}$ or $\{2, 7\}$, or with minimal discriminant at most 10^{14}. Assuming the paramodular conjecture, it contains a representative of every isogeny class of conductor at most 2^9.

The data (Chapter 6). The 512 curves consist of Smart's 366 curves with good reduction outside 2 and 146 further curves with bad reduction at an odd prime, where all of $3, 5, 7, 11, 13$ occur. Using Magma and Sage, and inspired by the invariants computed in the LMFDB, I computed for each curve a minimal model, field system, automorphism groups, conductor, torsion, 2-Selmer group, endomorphism algebra and endomorphism field, Sato–Tate group, rational points, analytic order of Sha, and the images of the mod-2 and mod-3 Galois representations, and identified the isogenies between Jacobians. For two curves where Chabauty's method failed, two-cover descent proved that there are no rational points. No curve with full rational 2-torsion was found; any such curve must have bad reduction at a prime $p \geq 17$. All curves and data are available online.

Future goals. A provably complete classification of all abelian surfaces over $\mathbb{Q}$ with good reduction away from $S = \{2\}$ remains open. The thesis points to some natural next steps: removing the principal polarisation assumption in Chapter 4 by working in GL_4, classifying isomorphism rather than isogeny classes, and determining the rational points on the remaining curves with quadratic or elliptic Chabauty. Completing the classification, by combining Galois-representation and L-function methods, is one of the projects in my research statement.